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Magnetic Polarity Sequence of the Upper Triassic Kayenta Formation

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ABSTRACT

The Kayenta Formation near Moab, Utah, contains four normal and four reversed intervals of polarity. These eight polarity intervals occur in approximately 100 m of medium- to fine-grained sediment. Because this type of sediment may indicate more rapid deposition, it appears that polarity changes were more frequent in Late Triassic time than in the Early Triassic. The mean pole for the Kayenta Formation is lat 61.9° N., long 74.4° E. ($\delta p = 3.7°$, $\delta m = 7.1°$), and lies on the western edge of other published Triassic poles. Poles for individual polarity zones are distributed in elongate fashion on a great circle, similar to distributions observed in other Triassic formations. This may indicate a long-term variation of the pole during Triassic time.

INTRODUCTION

The magnetization of the Upper Triassic Kayenta Formation has previously been studied by Runcorn (Nairn, 1956) and by Collinson and Runcorn (1960). These early results were scattered and included many directions near the present field direction; they therefore give inconclusive information about the Triassic field direction.

In 1972, we (Steiner and Helsley, 1972) reported preliminary results from a study of the magnetization of the Upper Triassic Kayenta Formation sampled on the Colorado Plateau in eastern Utah. We observed two sequences of samples of normal polarity, separated stratigraphically by 40 m of unsampled section. In addition, a third sequence, sampled at the base of the formation, showed unusual directions of magnetization, somewhat displaced from a typical Upper Triassic normal direction. In the uppermost sequence, three samples—one at the top and two at the base of the sampled interval—had reversed polarity. We have recently completed sampling the whole of the formation at the same locality, in order to determine whether such reversed zones were present and to define the entire polarity reversal sequence for the Kayenta Formation.

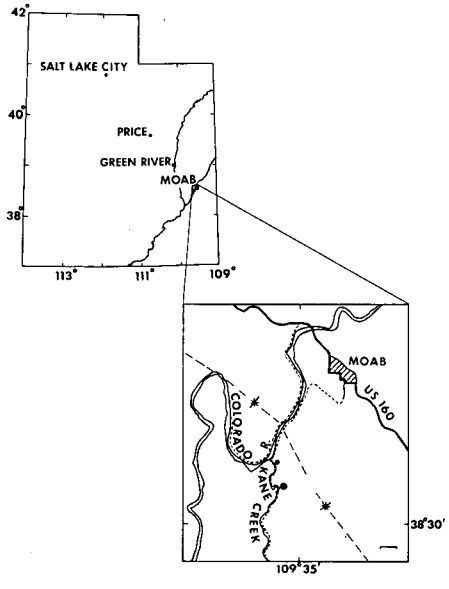


Figure 1. Sampling sites southwest of Moab, Utah. Main locality indicated by large dot; extra site slightly to the north indicated by small dot. Bar in lower right corner on lower diagram = 1 km.

GEOLOGIC SETTING

The fluvial beds of the Kayenta Formation were laid down conformably in most areas, on Wingate Sandstone in the uppermost part of Late Triassic time. The lower 30 m of the formation is a thickly bedded, cross-bedded, medium-grained sandstone, very similar lithologically to the upper part of the underlying Wingate Sandstone. The upper 70 m is predominantly a thinner bedded (although still cross-bedded) sandstone, with interbedded shale lenses. Local calcareous sandstone pebble conglomerates and mud chip conglomerates also occur. The upper section is generally somewhat finer grained and lighter in color than the lower part of the formation, having more variety of color (such as red-brown, white, chocolate brown, and lavender) and texture (coarse and fine sandstone, siltstone, shale, and conglomerate). The formation as a whole is coarser than the overlying and underlying formations.

Irregularity and lenticularity of bedding, cross-bedding, current lineation, channel fills, and mud chip conglomerates indicate the fluvial and probably fairly discontinuous nature of the deposition. Freshwater molluscs, reptile tracks, and plant remains indicate that the sediments were deposited in fresh water. The source of the sediment seems to have been to the northeast, probably derived from older sedimentary rocks uplifted along the western margin of Colorado.

The age of the Kayenta Formation is not well known and has been thought to be uppermost Late Triassic or Early Jurassic. However, recent reviews of the vertebrate fauna coupled with new vertebrate evidence generally favor an uppermost Triassic age (Lewis and others, 1961; Galton, 1971).

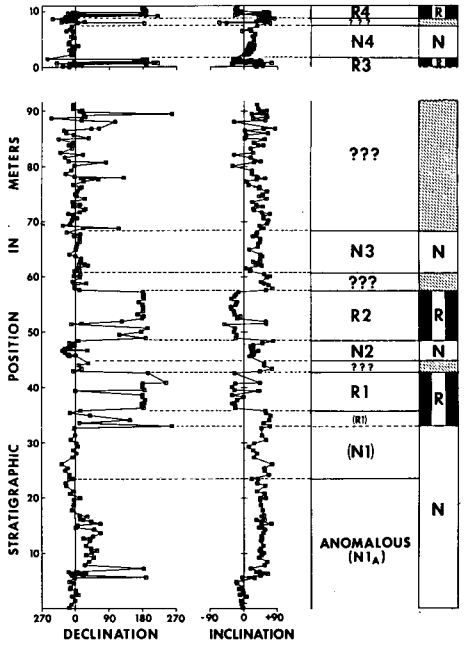


Figure 2. Stratigraphic plot of thermally demagnetized sample directions. Hachured intervals indicate sections of uncertain polarity.

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TECHNIQUES

All samples were obtained with a portable field drill and were individually oriented, stratigraphically discrete cores. All specimens were thermally demagnetized in sequential steps, in most cases to temperatures between 550° and 630°C; the residual field during cooling was ±5 γ. For most samples, the temperature at which a stable direction was reached is 550°C. However, in a few cases, a higher or lower temperature was used.

The optimum temperature at which the sample appeared to be free of secondary components was determined by a combined evaluation of the sample's behavior on orthogonal axes (Zijderveld) plots, stereographic plots, and logarithmic intensity plots. The temperature selected was that prior to which any of these plots showed peculiarities in sample behavior. Data were rejected if a marked change in sample direction or intensity then occurred. For weak samples, plots of the orthogonal axes were used: a marked deviation from a path toward the origin accompanied by failure of the next data set to return to this path determined the temperature selected.

LOCATION

Figure 1 shows the sampling location southwest of Moab, Utah. The locality is along the road through Kane Springs Canyon (lat 38.5° N., long 109.6° W.) and is an extension of the site of our earlier study (Steiner and Helsley, 1972). The entire formation has been sampled along the road and in natural outcrops between the road and the overlying Navajo Sandstone. However, a short distance north along the road (approximately a 1.6-km linear distance; Fig. 1), a thin-bedded sequence is preserved beneath the Navajo that was not observed at the first locality, and the uppermost 11 m below the Navajo Sandstone at this site was also sampled.

RESULTS

The lower 23 m of the formation has four groups of directions that are anomalous with respect to the Triassic normal direction (see Steiner and Helsley, 1972, 1974 [accompanying paper]): they most likely represent a period of normal polarity and will be referred to as $N1_A$. Above this interval, the magnetization is a sequence of well-defined normal and reversed directions. However, several intervals are not well defined; in part, these reflect the direction of the axial field (Fig. 2). The polarity intervals showing good directional resolution are numbered in ascending stratigraphic order.

Natural remanent magnetization (NRM) directions of R1, N2, R2, and N3 are shown in Figure 3; the thermally demagnetized directions are shown in Figure 4. A comparison of Figures 3 and 4 with Figure 2 shows that there is good polarity definition on demagnetization, because the NRM directions show little change during demagnetization.

In the interval between the anomalous directions and the first reversed zone, the directions are not well grouped, and a number of the samples have directions lying close to the axial field direction (D = 0°, I = 58°; Fig. 2). However, some of the samples in the lower part have typical normal Triassic directions, but the dispersion increases abruptly in the upper part of this interval. Directions are removed from this normal Triassic direction, some lying beyond the axial field direction toward a reversed Triassic direction. There is no difference between the lithologies of samples from this interval. We consider that the interval is normal in the lower portion (and part of N1) and succeeded by a few reversed samples in the upper part (lower part of R1).

The first reversed zone (R1) has an interesting characteristic: two samples in the middle of it in the stratigraphic sequence are normal (compare Figs. 2 and 4). These two samples do not change upon demagnetization and are nearly 180° away from the reversed directions. There are no lithologic differences between the two samples and the reversed samples stratigraphically above and below; all are a red calcareous sandstone conglomerate. The two samples could be interpreted as being a short polarity event, but more sampling is needed to confirm this. (A few normal samples are also present in R4, but they occur in a zone that may be weathered and thus may represent a somewhat later magnetization.)

The interval with high inclinations between R1 and N2 is probably part of R1 but is less magnetically stable because of its coarser, more friable lithology (relative to R1). Immediately above it the lithology changes to a dark-red siltstone, and the polarity changes to normal. The polarity of the narrow zone below N3 is uncertain. The directions cluster about the axial field direction, but a few move toward a normal Triassic direction upon demagnetization. The upper part of the formation also shows considerable instability. The upper boundary of N3 is drawn on the basis of the increase in scatter of directions and the steepening of the inclinations above this boundary. Many of the sample directions in the lower half of this interval above N3 move away from the axial field direction toward a normal direction upon demagnetization; thus, they may belong to N3. These directions are fairly well grouped. A few samples in the upper half of this interval move upon demagnetization toward a reversed direction and may indicate the existence of a reversed zone above. This upper half contains more dispersion than the lower half, but consecutive sample behavior is not consistent in either part,

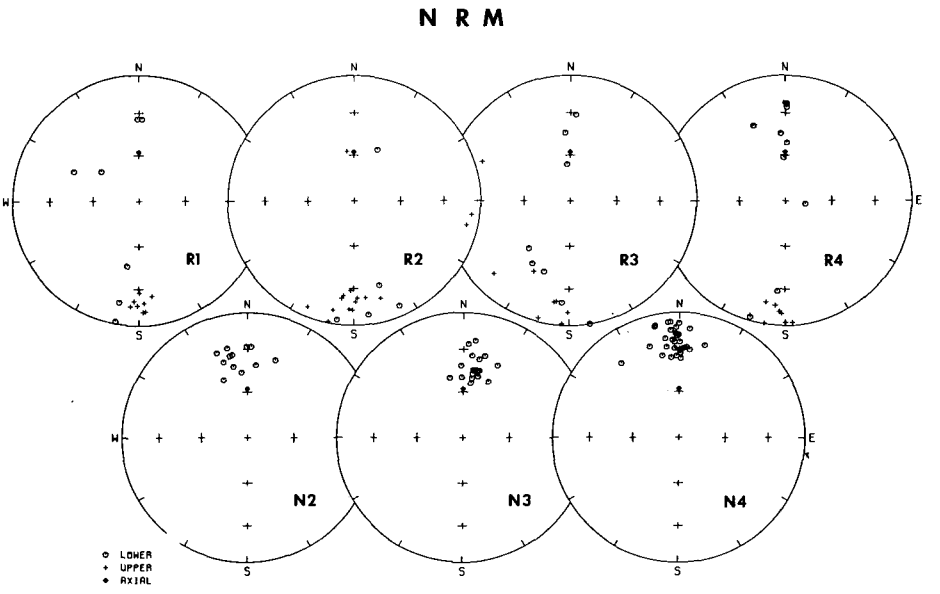


Figure 3. NRM directions of well-defined polarity zones.

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and the polarity of this interval is therefore uncertain.

As a result of sampling the upper 11 m immediately below the Navajo Sandstone 1.6 km north of the main locality, we observed three short polarity zones: reversed, normal, and reversed (R3, N4, R4 in Fig. 2). The NRM and demagnetized directions of these are also shown in Figures 3 and 4. As in the lower polarity zones, the NRM directions of these three polarity zones are also quite stable and show little change during demagnetization. A narrow interval of directions similar to that of the axial field, but containing one reversed direction, is present between N4 and R4. Another similar small zone occurs at the base of R3. Because our sampling was very dense at this site, we show this portion of the reversal sequence in expanded scale in Figure 5. The outcrops are discontinuous between the two sampling sites, which makes exact correlation between them slightly uncertain. The most reasonable correlation is that the lower reversed zone at the northern site (R3) is part of the unstable and possibly reversed zone at the top of the main site.

TABLE 1. DEMAGNETIZED MEAN DIRECTIONS OF THE KAYENTA FORMATION

	Declination (°)	Inclination (°)	No. of samples	Resultant R	Precision κ	α_{95}	Pole E. long	Pole N. lat	δp	δm
R1	181.5	-22.3	10	9.96	217.1	3.3	67.1	63.0	1.8	3.5
N2	346.9	19.9	11	10.53	21.4	10.1	96.4	59.4	5.5	10.6
R2	180.1	-20.7	15	14.82	76.1	4.4	70.3	62.2	2.4	4.6
N3	7.5	29.8	18	17.31	24.5	7.1	51.9	66.6	4.4	7.9
R3	189.2	-19.9	8	7.88	60.5	7.2	51.7	60.6	3.9	7.5
N4	352.3	13.3	30	29.50	58.6	3.5	84.8	57.5	1.8	3.5
R4	183.5	-15.0	10	9.94	161.8	3.8	63.7	59.0	2.0	3.9
By polarity	178.0	-20.3	7	6.93	81.0	6.8	74.4	61.9	3.7	7.1
All samples	178.9	-19.8	105	101.72	31.7	2.5	72.7	61.7	1.4	2.6

POLES

Pole positions were calculated for each normal or reversed zone (Table 1) assuming a Fisherian distribution of the data. All poles were calculated from the the mean direction. Diagram a in Figure 6 shows the NRM and demagnetized poles for each polarity zone. After demagnetization it is apparent that the polarity zone poles are distributed in an arc along a great circle: diagram b in Figure 6 illustrates this elongate distribution of poles.

The pole to the normal zone N3 is much more northerly than the rest of the demagnetized polarity zone poles. If its position is extrapolated further along the demagnetization path (which is directed away from the present pole), the pole would coincide with the great circle and would lie very near the reversed poles that are considered to be free of secondary components (a in Fig. 6). It is thus possible that N3 still contains some secondary components. There is a notable difference in the pole positions for N2 and N4, which are much more easterly than the poles for the reversed intervals and N3 (b in Fig. 6).

The mean pole position (Table 1) for the Kayenta Formation was calculated in two ways (using the same samples in both analyses): by giving unit weight to all samples with definite normal or reversed directions in the well-defined parts of polarity zones, and by giving unit weight to the mean direction of each polarity zone. Since the first calculation gives a deceptively small α_{95} due to the large number of samples, the second pole, 74.4° E., 61.9° N. (δp = 3.7, δm = 7.1), is preferred. This calculation is believed to be considerably more representative of the total formation and should replace poles given by us previously (Steiner and Helsley, 1972).

In diagram c in Figure 6, this mean pole is compared with other published Triassic results from North America. These results have been selected according to reliability criteria modified from Irving (1964). All undemagnetized results have been excluded, as well as poles lying very close to the axial dipole. Many of the eastern North American Upper Triassic igneous rocks have undergone zeolite facies metamorphism (Armstrong and Besancon, 1970), and many of the poles from these rocks are displaced toward the present or axial pole. Therefore, the poles may not represent the true field direction of that time and are not included in diagram c in Figure 6.

The mean pole for the Kayenta Formation lies on the western edge of the cluster of published demagnetized Triassic poles (c in Fig. 6). It is similar to the Upper Triassic Chinle Formation pole (Reeve and Helsley, 1972) but is distinctly different from the mean pole of the Lower Triassic Moenkopi Formation (Helsley and Steiner, 1974). The great circle fitting the polarity zone poles (b in Fig. 6) is similar to one fitting poles in the R4 interval of the Moenkopi Formation (Helsley and Steiner, 1974). (We have also observed this in the Chinle Formation.) At present the meaning of this elongate distribution of poles is unclear, but it may represent long-term variations of the pole during Triassic time (Helsley and Steiner, 1971, 1974). In turn, this may also be linked with the more westerly position of the mean Kayenta pole, because this pole, combined with many of the published Triassic poles, also forms an elongate distribution (c in Fig. 6).

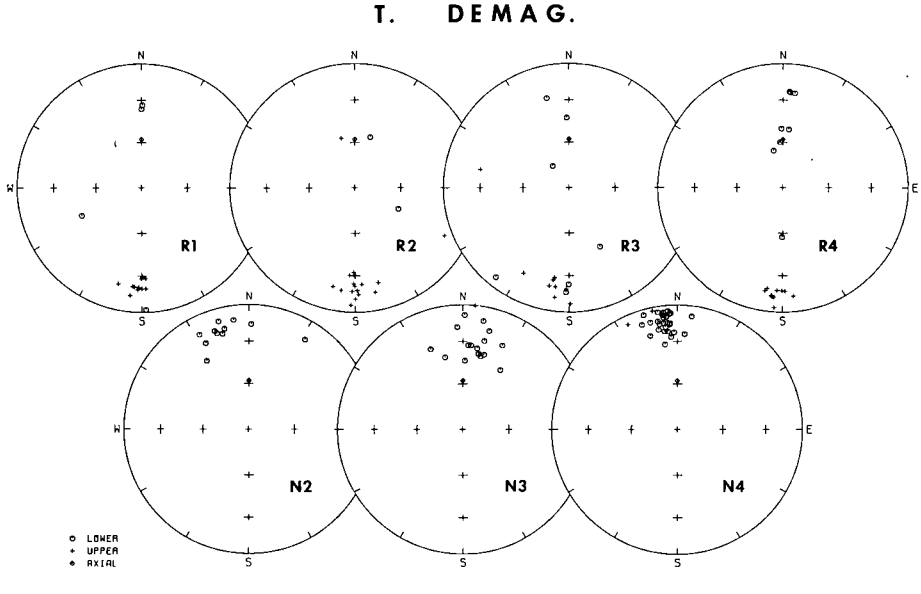


Figure 4. Thermally demagnetized directions of well-defined polarity zones.

CONCLUSIONS

The polarity sequence during Late Triassic Kayenta Formation time consisted of at least four normal and four reversed intervals of polarity. Striking features of this polarity sequence are the number and the short stratigraphic thickness of individual polarity zones when compared with other known Triassic polarity

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sequences. The Lower Triassic Moenkopi Formation (Helsley, 1969; Helsley and Steiner, 1974) and the Upper Triassic Chinle Formation (Reeve and Helsley, 1972) both have approximately the same number of polarity intervals as the Kayenta Formation (10 and 8, respectively). Yet, from published age assignments, the Moenkopi Formation likely spans a greater time interval than does the Kayenta Formation (Shoemaker and Newman, 1959). Furthermore, both the Moenkopi and Chinle Formations are finer grained depositional units than the Kayenta, and therefore they may represent greater amounts of time. In addition to representing a shorter time duration, deposition was somewhat irregular and discontinuous in the Kayenta, and lithologic boundaries are sometimes the polarity boundaries. Some portions of time are not represented in the Kayenta sediments, and more reversals may have occurred during this period of time than we have observed. Therefore, it is probable that the reversal frequency was significantly higher in latest Triassic time and was perhaps similar to that of the late Tertiary.

The pole positions of the polarity zones form an elongate distribution comparable to similar distributions in the Moenkopi and Chinle Formations. A mean pole, computed from individual polarity zone poles, occurs at lat 61.9° N., long 74.4° E. ($\delta p = 3.7°$, $\delta m = 7.1°$) and lies on the western edge of the distribution of published Triassic poles. The elongate distribution of the individual polarity zone poles may be another indication of long-term variations of the pole that occurred during Triassic time.

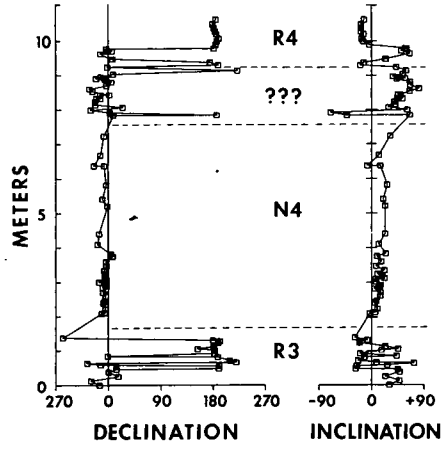


Figure 5. Expanded stratigraphic plot of top portion of section shown in Figure 2.

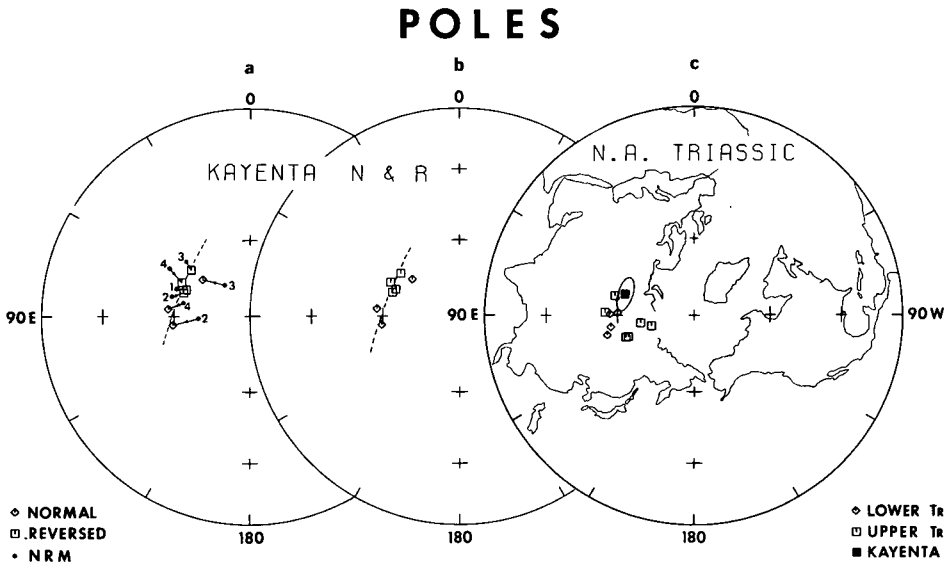


Figure 6. a, NRM and demagnetized poles of well-defined normal and reversed polarity zones; number by each NRM pole indicates polarity interval. b, Great circle fitting demagnetized poles of diagram a. c, Mean pole for Kayenta Formation with its α_{95} oval of confidence and published Triassic poles from North America (selection explained in text).

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